

Compass Hub

Installation guide for IT

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1. What you install

Compass Hub is the team service of Loonar Compass. It is one program, `compass-hub.exe`, without an installer and without Enterprise Architect. It runs as a Windows service and keeps:

- the team's EA models (read only): `.qea` / `.qeax` files or a PostgreSQL, SQL Server, MySQL or Oracle repository;
- the team policy, which the Compass apps follow: the minimum approval mode, web access, the AI only through the hub, and so on;
- the audit trail of the changes from all workstations, the change sets for review, and a web view for people without EA;
- the AI gateway: the Claude API keys stay on the server, with a key and limits per person or role;
- Jira and Confluence through one service account, a read-only GraphQL API, tables for Power BI and an MCP server.

Where	What	Network
Hub server	Service `CompassHub` (compass-hub.exe), listening only on `127.0.0.1:8730`	Local only
Hub server	IIS site `https://hub.company.com` with URL Rewrite + ARR in front of the service	Inbound 443 from the workstations
Identity provider	AD FS or Microsoft Entra ID: the "Compass Hub" application (OpenID Connect)	The browser of the user
Hub server → internet	`api.anthropic.com` (AI gateway), the Atlassian site (Jira/Confluence)	Outbound 443, only for the parts you use
Workstations	Loonar Compass (desktop and add-in in EA), settings forced by Group Policy	Outbound 443 to the hub

> This is the setup tested end to end on Windows Server 2022 with AD, AD CS, AD FS and IIS. Each step ends with a check, so a problem shows up at the step that caused it.

2. Before you start

✓	Item	Notes
☐	Windows Server 2019 or 2022, x64	2 vCPU, 4 GB RAM and 2 GB disk are enough for a team; more disk for large model files. No Enterprise Architect.
☐	DNS name, e.g. `hub.company.com`	Resolvable from the workstations.
☐	TLS certificate for that name	From the company CA or a public CA. **The workstations must trust its root** (Compass checks it against the Windows certificate store). The hub name must be in the certificate's SAN; from AD CS, the account that requests it needs Enroll on the Web Server template.
☐	IIS with the URL Rewrite and Application Request Routing (ARR) modules	Downloads from Microsoft (iis.net).
☐	AD FS (Windows Server) or Microsoft Entra ID	Only for company sign-in, which is recommended. Without it the hub has its own users and passwords.
☐	AD groups: `Compass-Admins`, `Compass-Architects`, `Compass-Viewers`	The group gives the role in the hub. Other names are possible (step 6).
☐	The models	.qea files copied to the server, or read access to the EA repository database (a read-only account is enough).
☐	Secrets ready, held by the person who installs	Claude API key(s), Atlassian API token, OIDC client secret, DB password. They are never put into hub.json.
☐	Outbound HTTPS from the server	`api.anthropic.com` (AI gateway), your Atlassian site (Jira/Confluence). Through a proxy: set `HTTPS_PROXY` for the service (step 3).

Files from the release: `CompassHub-0.9.3-windows-x64.zip` (with `compass-hub.exe`, `hub.example.json`, `README.md`, the licences), and for the workstations `LoonarCompass-Setup-0.9.3.exe` plus the `policies` folder (ADMX/ADML).

3. Step 1 – Files and folder

1. Check the SHA-256 of the download against the value published with the release:

   ```
   Get-FileHash .\CompassHub-0.9.3-windows-x64.zip -Algorithm SHA256
   ```

2. Unpack it to `C:\CompassHub` and create the subfolders `models` and `secrets`.
3. Restrict the folder to the service and the administrators. The hub writes its data here, and the secrets are here:

   ```
   icacls C:\CompassHub /inheritance:r /grant:r "SYSTEM:(OI)(CI)F" "BUILTIN\Administrators:(OI)
   (CI)F"
   ```

   On a Windows in another language, use the SID `*S-1-5-32-544` instead of the name of the Administrators group.

Check: `C:\CompassHub\compass-hub.exe --version` prints `0.9.3`.

4. Step 2 – hub.json and the models

Copy `hub.example.json` to `C:\CompassHub\hub.json`. A minimal start:

```
{
  "listen": "127.0.0.1:8730",
  "sources": [
    { "id": "architecture", "name": "Enterprise architecture", "properties":
"models/architecture.properties" }
  ],
  "policy": { "min_approval": "ask" }
}
```

- `listen` stays on `127.0.0.1`: only IIS on the same server talks to the hub.
- `sources`: one entry per model. `id` is short and stable, because the users and Power BI use it. `properties` is relative to hub.json.
- `readers` and `scopes` (in a source, optional): `"readers": ["anna", "role:architect"]` – only they (and admins) see the source at all; `"scopes": [{"for": "role:viewer", "packages": ["{GUID}", 42]}]` – the first entry for the person, their role or `*` limits them to those packages and their subpackages in the web view, search, element details, reports, Power BI tables and MCP. GraphQL, the assistant (Ask), impact and gap analysis and change sets cannot be limited by packages, so a person with a scope gets a clear refusal there.
- **Review and portfolio** (web view): under an element's details everybody who may read it comments or records a decision (ADR); an architect or the author resolves or accepts it (comments.json next to hub.json). The Portfolio tab counts the elements of a package by one or two properties (type, status, phase, author, a tagged value such as tag:Owner) as bars or a heat map, with PowerPoint and CSV.
- Leave out `users`: the hub then creates the account `admin` (step 4) and writes users back itself, as hashes only.

The model file, e.g. `C:\CompassHub\models\architecture.properties`:

```
# An EA project file (read only)
model.source=file
model.file=C:/CompassHub/models/Architecture.qea

# ... or a repository database (the password is the secret DB_PASSWORD):
# model.source=sqlserver          (or postgres, mysql, oracle)
# db.host=sql.company.com
# db.port=1433
# db.name=ea_repository
# db.user=compass_reader
# db.sslmode=verify-full
```

SQL Server, MySQL and Oracle are read through the vendor's 64-bit ODBC driver, which must be installed on the server. For SQL Server, "ODBC Driver 17 for SQL Server" (it comes with SQL Server 2022) or 18 is enough; the installed drivers: `Get-OdbcDriver -Name '*SQL Server*' -Platform 64-bit`. PostgreSQL and files need nothing more. Every read is rolled back, so a read-only database account is enough. SQL Server without the secret `DB_PASSWORD` uses Windows authentication: the hub service (LocalSystem) signs in as the computer account (`DOMAIN\SERVER$`), which then needs the `db_datareader` role in the model database.

- **Named instance** (e.g. SQL Server Express): `db.host=SERVER\SQLEXPRESS` without a port, with the SQL Server Browser service running; or a fixed TCP port (SQL Server Configuration Manager → TCP/IP → IPAll: TcpPort 1433, TcpDynamicPorts empty) and `db.host` / `db.port`.
- **Your own connection string:** `db.odbc=Driver={ODBC Driver 17 for SQL Server};Server=sql.company.com,1433;Database=ea_repository;Trusted_Connection=Yes` replaces host, port, name and user (without a password; `DB_PASSWORD`, if set, is added).
- **db.sslmode** for SQL Server: `prefer` = encryption when the server and the driver use it (no certificate check); `require` = always encrypted, the certificate is not checked; `verify-full` = encrypted and the certificate must be trusted and match the server name (not with a self-signed SQL Server certificate).

> Keep the hub's model files separate from the files people edit in EA. To refresh a file model, replace the file. The hub opens it read-only for each request, so no restart is needed.

5. Step 3 – Secrets

Each secret is a text file in `C:\CompassHub\secrets` that contains only the value. The service learns the path from an environment variable `LOONAR_SECRET_<NAME>_FILE`. The person who knows the value creates the file, e.g. in Notepad or with:

```
Read-Host -AsSecureString "Value" | ConvertFrom-SecureString -AsPlainText | Set-Content
C:\CompassHub\secrets\claude_key_team -NoNewline
```

(`-AsPlainText` needs PowerShell 7; in Windows PowerShell 5.1, Notepad is simpler.)

File (example)	Variable for the service	Needed for
`claude_key_team`	`LOONAR_SECRET_CLAUDE_KEY_TEAM_FILE`	The AI gateway: one variable per key named in hub.json (step 7)
`claude_api_key`	`LOONAR_SECRET_CLAUDE_API_KEY_FILE`	"Ask" in the web view
`jira_token`	`LOONAR_SECRET_JIRA_TOKEN_FILE`	Jira
`confluence_token`	`LOONAR_SECRET_CONFLUENCE_TOKEN_FILE`	Confluence
`oidc_client_secret`	`LOONAR_SECRET_OIDC_CLIENT_SECRET_FILE`	Company sign-in
`db_password`	`LOONAR_SECRET_DB_PASSWORD_FILE`	A model database

The name in the variable is the secret name in capitals, with other characters as `_`: the hub.json secret `claude_key_anna` is `LOONAR_SECRET_CLAUDE_KEY_ANNA_FILE`.

The variables are set on **the service only**, not for the whole computer, after the service is installed (step 4). This takes effect at the next start of the service, with no reboot:

```
$vars = @(
  'LOONAR_SECRET_CLAUDE_KEY_TEAM_FILE=C:\CompassHub\secrets\claude_key_team',
  'LOONAR_SECRET_JIRA_TOKEN_FILE=C:\CompassHub\secrets\jira_token',
  'LOONAR_SECRET_CONFLUENCE_TOKEN_FILE=C:\CompassHub\secrets\confluence_token',
  'LOONAR_SECRET_OIDC_CLIENT_SECRET_FILE=C:\CompassHub\secrets\oidc_client_secret'
  # 'HTTPS_PROXY=http://proxy.company.com:8080'   only if the server goes out through a proxy
)
Set-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\CompassHub -Name Environment -Type
MultiString -Value $vars
Restart-Service CompassHub
```

The hub reads a secret when it first needs it. A missing one is reported where it is used, e.g. "secret 'jira_token' is not set: give it as LOONAR_SECRET_JIRA_TOKEN (or a file named in LOONAR_SECRET_JIRA_TOKEN_FILE)". The values never appear in hub.json, the logs or the web view. A byte order mark at the start and spaces or a new line at the end of a secret file (Notepad) are ignored.

6. Step 4 – The Windows service and the first sign-in

1. In PowerShell as administrator:

   ```
   C:\CompassHub\compass-hub.exe --service-install C:\CompassHub\hub.json
   ```

   This creates the service `CompassHub` ("Compass Hub (Loonar Compass team service)"), with automatic start under LocalSystem, and starts it.
2. Now set the secrets of the service (step 3) and run `Restart-Service CompassHub`.
3. On the server, open `http://127.0.0.1:8730/` and sign in as `admin` / `passw0rd` (with a zero). The hub asks for a new password at once and allows nothing else until then. The password needs at least 10 characters. Do this **before** step 5 makes the hub reachable. If `hub.json` already has a company sign-in (`oidc`), there is no `admin` account: sign in with an account from the admin group (`roles.admin`).

Service account (recommended for production). LocalSystem is the simplest start, but whoever took over the hub would act as SYSTEM, and Windows authentication to SQL Server uses the computer account. Better: a dedicated domain account or a group managed service account (gMSA) with only the rights the hub needs – modify on `C:\CompassHub`, read on the models, `db_datareader` in the model databases: `sc.exe config CompassHub obj= "DOMAIN\svc-compasshub$"` (gMSA, no password) and `Restart-Service CompassHub`.

Check:

- `Get-Service CompassHub` shows Running;
- `C:\CompassHub\hub.log` shows the start and your sign-in;
- the web view lists the model from step 2.

7. Step 5 – HTTPS with IIS (URL Rewrite + ARR)

The hub speaks plain HTTP on 127.0.0.1. IIS gives it the company name and certificate.

1. Install IIS with URL Rewrite and ARR. Create a site, e.g. **CompassHub**, with a binding for `https`, port 443, the host name `hub.company.com` and the certificate. The physical path is an empty folder, e.g. `C:\inetpub\compasshub`. If the hub has no company sign-in (`oidc`), add `"allowed_hosts": ["hub.company.com"]` to `hub.json`: a hub on 127.0.0.1 answers only to its own names (otherwise 421).
2. Turn on the proxy and switch off rewriting the host in redirects. This is **required**; otherwise ARR rewrites the redirect to AD FS to the hub's own address, and the company sign-in fails. Set the timeout for long AI answers as well:

   ```
   Import-Module WebAdministration
   $p = 'MACHINE/WEBROOT/APPHOST'
   Set-WebConfigurationProperty -PSPath $p -Filter system.webServer/proxy -Name enabled -Value True
   Set-WebConfigurationProperty -PSPath $p -Filter system.webServer/proxy -Name
   reverseRewriteHostInResponseHeaders -Value False
   Set-WebConfigurationProperty -PSPath $p -Filter system.webServer/proxy -Name timeout -Value
   '00:05:00'
   ```

3. Put this `web.config` in the site folder. It forwards everything to the hub and allows request bodies up to 32 MB, because questions with model diagrams are large:

```
<?xml version="1.0" encoding="utf-8"?>
<configuration>
  <system.webServer>
    <rewrite>
      <rules>
        <rule name="CompassHub" stopProcessing="true">
          <match url="(.*)" />
          <action type="Rewrite" url="http://127.0.0.1:8730/{R:1}" />
        </rule>
      </rules>
    </rewrite>
    <security>
      <requestFiltering>
        <requestLimits maxAllowedContentLength="33554432" />
      </requestFiltering>
    </security>
  </system.webServer>
</configuration>
```

4. Firewall: allow inbound 443 only. Port 8730 is never opened, because the hub listens on 127.0.0.1.
5. AD FS on the same server: AD FS also uses port 443 (HTTP.sys). Tick **Require Server Name Indication** on the IIS binding (`New-WebBinding … -SslFlags 1`), or IIS and AD FS get in each other's way.
6. The hub does not need `preserveHostHeader` or `X-Forwarded-Proto`: its addresses come from `redirect_uri` in hub.json.

The hub trusts `X-Forwarded-For` (which ARR adds) only from a proxy on the same computer. The audit trail and the lockout of wrong tokens then see the real client address.

Check: from a workstation, `https://hub.company.com/` shows the sign-in page with no certificate warning.

8. Step 6 – Company sign-in (AD FS or Entra ID)

AD FS

1. AD FS Management → Application Groups → Add, then the template **Server application accessing a web API**, name "Compass Hub".
2. Server application:
   - note the **client identifier**;
   - redirect URI: `https://hub.company.com/auth/callback`;
   - credentials: **Generate a shared secret**. Save it right away into `C:\CompassHub\secrets\oidc_client_secret`; AD FS shows it only once.
3. Web API:
   - identifier: the client identifier from the previous step;
   - access control policy: the people who may use the hub, e.g. "Permit everyone" (roles still come from the groups);
   - permitted scopes: `openid`, `profile`, `email`, `allatclaims`.
4. Web API → Issuance Transform Rules → "Send LDAP Attributes as Claims" (Active Directory):
   - *User-Principal-Name* → `upn`;
   - *E-Mail-Addresses* → `email`;
   - *Display-Name* → `name`;
   - *Token-Groups - Unqualified Names* → `groups`.

   (Type the outgoing claim names exactly like that.)

In `hub.json` (the hub adds `allatclaims` for AD FS itself, so the claims reach the ID token):

```
"oidc": {
  "issuer": "https://adfs.company.com/adfs",
  "client_id": "<client identifier>",
  "redirect_uri": "https://hub.company.com/auth/callback",
  "groups_claim": "groups",
  "roles": { "admin": ["Compass-Admins"], "architect": ["Compass-Architects"], "viewer": ["Compass-
Viewers"] },
  "password_login": true
}
```

Microsoft Entra ID

The setup in Entra ID:

- **App registration:** platform *Web*, redirect URI `https://hub.company.com/auth/callback`, a client secret (into the secret file), and the ID token with the optional claims `upn` and `email`.
- **Issuer:** `https://login.microsoftonline.com/<tenant-id>/v2.0`.
- **Groups:** Entra sends them as object IDs by default. Either put those IDs in `roles`, or, for groups synchronised from AD, choose "sAMAccountName" in the token configuration.

Rules the hub applies

- The role comes from the groups at every sign-in, and the highest one wins. A person in none of the groups gets no access.
- A company account with the same name as a local hub user is refused until the local user is renamed or removed.
- The hub checks the ID token itself: the RS256 signature, issuer, audience, expiry and nonce, with PKCE and a one-time state.
- When everything works, set `"password_login": false`: only the company button remains. API tokens for MCP clients still work. Keep one local admin password in your vault, for the case that the identity provider is down.

Check: after `Restart-Service CompassHub`, "Sign in with your company account" leads to AD FS and back. hub.log shows "… signed in with a company account as architect", and the **Users** tab lists the company account with its role.

9. Step 7 – AI keys, Jira, Confluence and the team policy

Add the sections to `hub.json`. Every secret named here also needs its `…_FILE` variable (step 3). Then run `Restart-Service CompassHub`; the signed-in people stay signed in.

```
"ai": {
  "keys":   [ { "for": "role:architect", "secret": "claude_key_team" },
              { "for": "anna@company.com", "secret": "claude_key_anna" } ],
  "limits": [ { "for": "*", "tokens_per_day": 2000000, "tokens_per_month": 30000000 } ],
  "models": [ "claude-haiku-5-5", "claude-sonnet-5-5" ]
},
"integrations": {
  "jira":       { "url": "https://company.atlassian.net", "user": "svc-compass@company.com",
"project": "ARCH", "issue_type": "Task" },
  "confluence": { "url": "https://company.atlassian.net/wiki", "user": "svc-compass@company.com",
"space": "ARCH" }
},
"policy": {
  "min_approval": "ask",
  "ai_via_hub": true,
  "jira_via_hub": true,
  "confluence_via_hub": true,
  "web_access": false,
  "update_check": false
}
```

Setting	Meaning
`ai.keys`	Which Claude key a person uses. Order: the user name, then `role:<role>`, then `*`. No match means no AI for that person.
`ai.limits`	Tokens per day and per month, and also USD per day and per month (`usd_per_day`, `usd_per_month`: the cost at the model's prices, cache included). Above the limit, the hub refuses with a clear message. The usage summary shows USD today and this month.
`ai.models`	The allowed models; other models are refused.
`ai.openai`	The gateway for OpenAI-compatible models: `{"url": "" (OpenAI) or an Azure OpenAI / other base, "keys": [{"for": …, "secret": …}]}`. Compass with the provider OpenAI and `policy.ai_via_hub` sends its requests here; the same limits, budgets and allowed models apply.
`policy.ai_via_hub`	Compass asks the AI only through the hub: no API key on the workstations.
`policy.jira_via_hub`, `confluence_via_hub`	Jira/Confluence through the service account of the hub. A viewer may only read; every change is audited.
`policy.min_approval`	The weakest approval mode allowed in Compass for writes to the model (`ask` is the strictest).
`policy.web_access`, `web_allowed_domains`	Whether the assistant may search the web or read pages, and from which domains.
`policy.update_check`	Whether Compass checks GitHub every hour for a new version.

Admins change the policy later in the **Team policy** tab of the web view, and the usage per person is in **Users** → **AI usage**, together with the cost in USD (today and this month) by model and by role and a **Download CSV** button.

10. Step 8 – The workstations (Group Policy)

1. **Trust the certificate.** If the hub certificate comes from the company CA, deploy its root to the workstations (GPO: Computer Configuration → Policies → Windows Settings → Security Settings → Public Key Policies → Trusted Root Certification Authorities). Without it, Compass reports an untrusted certificate (UnknownIssuer).
2. **Templates.** Copy `LoonarCompass.admx` to `\\<domain>\SYSVOL\<domain>\Policies\PolicyDefinitions`, and `en-US\LoonarCompass.adml` (and `pl-PL` if needed) to the language subfolders. Without a central store, copy them to `C:\Windows\PolicyDefinitions` on the computer where you edit the GPO.
3. **GPO** for the workstation OU: Computer Configuration → Administrative Templates → Loonar → Loonar Compass.

Policy	Registry value (HKLM\SOFTWARE\Policies\Loonar\Compass)	Effect
Hub address	`HubUrl` (text)	The hub address is fixed in Compass and cannot be changed.
Require the hub	`RequireHub` = 1	The assistant works only while the person is signed in to the hub.
Web access	`WebAccess` = 0	No web search and no reading of pages; the 🌐 button is disabled.
Update check	`UpdateCheck` = 0	No hourly check for a new version.
Minimum approval mode	`MinApproval` = ask / plan / auto	A weaker mode cannot be chosen.
Allowed AI providers	`AllowedProviders`, e.g. `anthropic`	Other providers (e.g. OpenAI) are refused.

Without a domain, the same values can be imported with `LoonarCompass-company-policy.reg` (and removed with `LoonarCompass-remove-policy.reg`). The computer policy is stronger than the hub policy and the user's settings; it can only make Compass stricter.

Installing Compass. `LoonarCompass-Setup-0.9.3.exe` installs for the current user (`%LOCALAPPDATA%\Programs\Loonar Compass`) or for all users (`Program Files`), and registers the add-in in Enterprise Architect. For deployment tools (Intune, Configuration Manager, a startup script) it installs without a window:

```
LoonarCompass-Setup-0.9.3.exe --quiet --all-users
```

Option	Effect
`--quiet`	No window and no questions. Exit code 0 = installed, 1 = error, 740 = `--all-users` without administrator rights. The result is in `%TEMP%\LoonarCompass-Setup.log` (for SYSTEM: `C:\Windows\Temp`).
`--all-users`	For everybody: Program Files, the common Start menu, the add-in in EA for all users (HKLM). Run it as SYSTEM or as an administrator.
`--dir <folder>`	Another program folder.
`--lang en|pl|de|fr`	The language of the licence and changelog files.
`--no-desktop`, `--no-start-menu`	Without these shortcuts.
`--no-ea-addin`	Without the add-in in Enterprise Architect.

The same command updates an existing installation. Close Compass and EA on the computer first. The user settings are not touched; with the computer policy (hub address, required hub) nobody needs to configure anything. Quiet removal: `"C:\Program Files\Loonar Compass\uninstall.exe" --uninstall --quiet`.

The user's first sign-in. In Compass, open Settings → Team (Compass Hub) → **Sign in with your company account**. The browser opens AD FS and returns to Compass. The sign-in lasts 12 hours and survives a restart of the hub.

11. Step 9 – Acceptance checklist

✓	Test	Expected
☐	Open `https://hub.company.com/` from a workstation	The sign-in page, no certificate warning
☐	Company sign-in as a member of Compass-Architects	Role architect; hub.log "signed in with a company account as architect"
☐	A person in none of the Compass groups	Refused
☐	Compass with the GPO, before the sign-in: send a question	"Your organisation requires the team hub: sign in to it in Settings → Team (Compass Hub)..."
☐	The same after the sign-in	An answer, with no API key on the workstation; a row in Users → AI usage
☐	A viewer asks a question (no key for viewers)	"the hub has no AI key for you"
☐	The 🌐 button in Compass	Disabled: "switched off by your organisation"
☐	`Restart-Service CompassHub`, then ask again	Works without signing in again
☐	Hub stopped, ask	A short message that the hub does not answer; EA keeps working
☐	Jira: "Create a Jira issue: test" (architect)	The issue in the project; an entry in the hub audit trail with the person's name

12. Operation: update, backup, logs, removal

Update

```
Stop-Service CompassHub
Copy-Item .\compass-hub.exe C:\CompassHub\compass-hub.exe -Force   # from the new zip, after checking
its SHA-256
Start-Service CompassHub
```

hub.json, the secrets and the sign-ins stay as they are. People remain signed in.

What to back up (C:\CompassHub)

File / folder	Content
`hub.json`	Configuration, local users (password and token hashes only), policy
`audit.jsonl`	The audit trail of changes and sign-ins
`changes\`, `notes\`	Change sets for review and the team notes
`ai_usage.jsonl`	AI usage per person (the base of the limits)
`company_accounts.json`, `sessions.json`	Company accounts seen and current sign-ins (hashes); can be lost without harm: people sign in again
`secrets\`	Only if your backup is encrypted. Otherwise keep the values in the password vault.
`models\`	The model files, if they are not kept elsewhere

Logs

- `hub.log`: start, sign-ins, refused requests, errors. An "internal error (id ...)" shown to a user has its details here under the same id.
- `audit.jsonl` and `hub.log` are renamed with a date when they grow large.

Removal

```
C:\CompassHub\compass-hub.exe --service-uninstall
```

Then remove the IIS site, the AD FS application group and the GPO, and delete `C:\CompassHub` after the backup.

13. Troubleshooting

Symptom	Cause and fix
Compass: certificate not trusted / UnknownIssuer; the browser shows "Not secure"	The workstation does not trust the root of the hub certificate. Deploy the root CA (step 8), then restart the browser and Compass.
Company sign-in ends at `https://hub.company.com/adfs/…` or with 404	ARR rewrites the redirect. Set `reverseRewriteHostInResponseHeaders` = False (step 5).
"<name> is not in any group of the hub (…): ask the administrator for access"	The person is in none of the role groups, or the `groups` claim is missing. Check the transform rule "Token-Groups - Unqualified Names → groups" and the `allatclaims` scope.
"A local user of the hub has the same name"	Rename or remove that local user in the Users tab.
Compass: "the team hub or the server in front of it does not answer properly (502)"	The service is stopped or starting: `Get-Service CompassHub`, then hub.log.
Compass: "your sign-in to the team hub has expired"	12 hours have passed, or sessions.json was lost. Sign in again in Settings → Team.
"the hub has no AI key for you"	No `ai.keys` entry for the person, their role or `*`.
"your AI limit for today is used up"	Raise `ai.limits` for the person or role, or wait for the next day.
"secret '…' is not set: give it as LOONAR_SECRET_…"	The `…_FILE` variable is missing in the service's `Environment`, or the file is empty or not readable by SYSTEM.
hub.log: "<address>: 20 wrong tokens, refused for 10 minutes"	20 **different** wrong tokens came from one address (an old token repeated counts once). Valid tokens still work. Find the client in hub.log.
Large questions fail with 413 or 404.13	`maxAllowedContentLength` in web.config (step 5).
Long answers end with 502.3 after 2 minutes	The ARR `timeout` (step 5).
AD FS: MSIS7628 from `Grant-AdfsApplicationPermission` right after installing AD FS	The farm is still starting: repeat the command a few minutes later.
hub.log / AD FS: MSIS9622 "Client authentication failed"	The secret file does not match AD FS. Make a new secret: `Set-AdfsServerApplication -TargetIdentifier <client id> -ResetClientSecret`, write it to the file, `Restart-Service CompassHub`.

Compass on a Windows Server (e.g. Remote Desktop Services): PDF export fails, or the first Office sign-in is blocked	Compass makes PDFs with Word and PowerPoint on the computer where it runs. On Windows Server, IE Enhanced Security Configuration can block the first Office sign-in (`odc.officeapps.live.com`): add the Office sign-in sites to the trusted sites, or turn the setting off for users in Server Manager.

14. Linux and Docker

The same hub runs in a container. The data folder (hub.json with `"listen": "0.0.0.0:8730"`, the models and the hub's files) is mounted as `/data` and must be writable by user 10001. The secrets come from `LOONAR_SECRET_*` variables or from Docker secrets through `LOONAR_SECRET_*_FILE`:

```
docker run -d --name compass-hub -p 127.0.0.1:8730:8730 -v /srv/compass-hub:/data \
  -e LOONAR_SECRET_CLAUDE_KEY_TEAM_FILE=/run/secrets/claude_key_team \
  ghcr.io/loonar-eu/compass-hub:0.9.3
```

Put nginx or another HTTPS proxy in front of it, as in step 5. The proxy must keep the `Location` header of redirects unchanged, pass `X-Forwarded-For`, allow 32 MB bodies and a 5-minute timeout. For SQL Server, MySQL or Oracle models, build your own image on this one and add the vendor's Linux ODBC driver.